



AI-Powered Document Verification

Stop fake pay stubs, diplomas, and professional certs **before they close the deal.**

VICTIG's Document Fraud Detector combines forensic metadata analysis, mathematical validation, tax-form structure checks, cross-document consistency analysis, and Claude AI vision to catch fabricated employment, income, and education documents that slip past manual review — in seconds, not hours. Every rule has been tuned against real fraud caught in production verification work.

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[See How It Works →](#)

8

DOCUMENT TYPES
SUPPORTED

9

DETECTION LAYERS

< 30s

AVERAGE ANALYSIS
TIME

100%

DOCUMENTS PURGED
AFTER REVIEW

THE PROBLEM

Fake documents are getting easier — and better.

AI generators, editable templates, and \$5 online forgery services have made it trivial for candidates to fabricate the exact documents employers rely on for verification. Manual review can't keep up.



AI-Generated Documents

ChatGPT and specialized tools now produce pay stubs and W-2s indistinguishable to the naked eye — but not to forensic metadata.



Photoshop & Canva Edits

Salary numbers, employment dates, and employer names are altered on real documents. Traditional review misses these edits.



Template Fabrication

Fill-in-the-blank pay stub templates from paystubcreators.com and similar sites create "documents" that look official but never came from a real payroll system.

CAPABILITIES

Nine layers of detection working together.

Any single check can be beaten. Combining metadata forensics, math validation, content analysis, visual and color forensics, tax-form structure checks, cross-document consistency analysis, credential intelligence, and AI vision produces a fraud signal that's remarkably hard to fake.



Metadata Forensics

Every file carries a digital fingerprint. We read it before anyone reads the document.

- ✓ Creation and modification timestamps — detects the "January pay stub" created yesterday, files modified after creation, and documents created the same day they're submitted
- ✓ Author, creator, and producing software — 25+ image editors, PDF editors, and office tools risk-weighted individually (Photoshop ≠ Microsoft Word ≠ Acrobat)
- ✓ Human-authored metadata tells — e.g. a PDF /Subject reading "Employment verification" on a form that should come straight from a payroll system
- ✓ Redaction-aware scoring: when a candidate legitimately redacted PII, editing-software penalties are automatically reduced instead of crying wolf
- ✓ File-type spoof detection (.pdf extension on a non-PDF file)
- ✓ Documents from 20+ recognized payroll systems earn a *negative* risk adjustment — legitimate sources are rewarded, not just fraud punished



Mathematical Validation

Fraudsters change numbers without understanding tax law. Math doesn't lie.

- ✓ Gross – deductions = net (with tolerance) — a net that exceeds gross is an instant critical flag
- ✓ Hours × rate = gross pay reconciliation
- ✓ Social Security tax at 6.2%, Medicare at 1.45%, impossible federal-withholding rates
- ✓ Deduction ratios outside the realistic 15–45% band — suspiciously low *and* suspiciously high
- ✓ W-2 Box 1 / 3 / 5 wage consistency
- ✓ Implausibly low wages carrying tax withholding (sub-\$2,000 W-2 wages with federal/state withholding — real workers at that income are generally exempt)
- ✓ Missing two-digit cents on W-2 dollar amounts — IRS Pub. 1141 requires them; word-processor forgeries almost always omit them
- ✓ Tax years in the future (impossible) or issued before year-end (premature)



Text & Content Analysis

Templated documents leave verbal fingerprints — if you know where to look.

- ✓ Placeholder text ("Lorem ipsum", "John Doe", "[Company Name]")
- ✓ Invalid field values — "N/A", "NOT NEEDED", "None" in boxes real payroll software fills with \$0.00 or leaves blank
- ✓ EIN intelligence: format validity, geographic mismatch between EIN prefix and employer state, and high-risk online-application prefixes
- ✓ W-2 Box 14 code validation against the standard payroll code vocabulary (random strings like "OB3QOT" get flagged)
- ✓ Employer-name heuristics: nonsense names, improper capitalization ("yinzer rentals"), and bulk e-filing platform fingerprints with system-generated account numbers
- ✓ Multiple W-2 forms crammed on a single page (multiple EINs/employer blocks)
- ✓ State wages/tax present with a blank Box 15 employer state ID — fabricators rarely know the state registration number
- ✓ Inconsistent date formats, suspicious phrases like "proof of income" that real payroll never prints
- ✓ SSN truncation compliance with IRS Pub. 1586 — full-SSN redaction is treated as candidate privacy protection, not fraud



Visual & Color Forensics

Cut-and-paste operations leave traces invisible to the naked eye.

- ✓ Resolution, aspect ratio, and color-profile checks; standard paper-size validation
- ✓ Cropped/truncated image detection — extreme aspect ratios that hide part of a form
- ✓ Non-black machine text: real tax documents print in black; blue/red typeset field data is a template-overlay tell — with automatic white-balance correction so a warm-lit phone photo of real black ink doesn't false-positive
- ✓ Bimodal text-darkness patterns — two ink populations on one page means overlaid or pasted text
- ✓ Font-size clustering across value fields (computer-vision backstop, corroboration-gated so clean scans don't trip it)
- ✓ Edge artifacts, noise-pattern inconsistencies, screenshot-of-a-screenshot detection
- ✓ Physical marker/ink redaction recognition — expected candidate privacy behavior, scored accordingly
- ✓ Anomaly compounding: several individually minor oddities on one document escalate together



Tax-Form Structure Forensics

The IRS publishes exact form layouts. Fabricated W-2s and 1099s get the structure wrong in ways real payroll software never does.

- ✓ W-2 Box 12 must be four rows labeled 12a–12d; repeated unlabeled "12/12/12" rows expose a homemade template
- ✓ W-2 Box 13 is one box with three checkboxes — three rows each labeled "13" is a fabrication tell
- ✓ Duplicated or out-of-sequence box numbers vs. the official IRS layout
- ✓ On genuine W-2s/1099s the tax year prints larger and bolder than body text — same-size years suggest a template
- ✓ Form rule-lines crossing or clipping through text, and text colliding with checkboxes or bleeding between fields
- ✓ Missing or inconsistent box borders in the state-tax row
- ✓ Years or amounts overprinted on top of covered content (white/black boxes with new text overlaid)



IRS Transcript Cross-Document Analysis

IRS Wage & Income Transcripts are only released as uneditable PDFs. Re-flowed, edited copies leak structural tells — and comparing a batch exposes what single-document review never sees.

- ✓ Batch mode: multiple transcripts analyzed together, cross-checked for impossible combinations
- ✓ Duplicate tracking numbers across "separate" transcripts — a smoking gun; genuine packets are one continuous document
- ✓ Same tracking number with different PDF creation times or page dimensions
- ✓ Missing IRS masthead, sensitive-data banner, tracking number, or response date
- ✓ Future-dated request/response dates
- ✓ Truncated system values ("Origin" instead of "Original", "Un" instead of "Unanswered") left behind when a transcript is re-typed into an editable format
- ✓ Replacement footers like "NON-EMPLOYMENT INFORMATION REDACTED" — real IRS output never prints this
- ✓ Employer-name truncation analysis based on the IRS's actual 3–4 character word-group convention — including over-clipped names no IRS system would emit
- ✓ Solid black bars over wage columns detected via pixel-run analysis



Education Credential Verification

Diplomas and transcripts are increasingly fabricated — our engine catches the tells.

- ✓ Documented diploma-mill blocklist — 60+ institutions compiled from GAO reports, the Oregon ODA unaccredited list, and academic-integrity research
- ✓ Diploma-mill language detection: "life experience degree", "no attendance required", "degree in one day", "instant degree", vague "internationally accredited" claims with no named accreditor
- ✓ Transcript GPA reconciliation: cumulative GPA re-computed from every listed course, grade, and credit hour — small drift is tolerated, large gaps escalate
- ✓ Credit-hour audits: stated totals vs. summed course credits, per-course loads outside the US norm, improbable all-A transcripts, GPAs beyond the 4.0/4.3 scale
- ✓ Latin honors thresholds (cum laude ≥ 3.5 , magna ≥ 3.7 , summa ≥ 3.9) cross-checked against the printed GPA
- ✓ Future-dated or implausibly old conferral years; degree names outside standard US nomenclature; diplomas with no identifiable institution
- ✓ US Dept of Ed recognized accreditor detection (HLC, MSCHE, WSCUC, ABET, LCME, AACSB, ABA, CCNE/ACEN...) as a positive signal
- ✓ Legitimate source recognition: Parchment, National Student Clearinghouse, MyCreds, Ellucian Banner, Workday Student, PeopleSoft Campus
- ✓ Credential misspelling blocklist ("university", "colledge", "cume laude", "magma cum laude", "curriculum", "diplomma") — real registrars don't misspell their own vocabulary
- ✓ AI vision review of seals (embossed vs. pasted), signatures and titles, course-code consistency, realistic semester loads, and template-vs-professional-print quality



Professional License / Certification Checks

Fake CPA, PMP, and licensing certs are a growing verification vector.

- ✓ 30+ recognized issuing bodies: AICPA/NASBA, PMI, ISACA, (ISC)², NCEES, SANS/GIAC, CompTIA, AWS/Microsoft/Google/Cisco, SHRM/HRCI, ANCC, NABP, NBCOT, APTA, ASHA, medical specialty boards, state bars and licensing boards
- ✓ Fabricated-issuer flagging — fake certs love vague, invented "institutes" and "boards"
- ✓ Credential-number sanity: too-short IDs, repeated-character and placeholder patterns ("12345", "XXXXX")
- ✓ Expiration plausibility: long-expired credentials, and "lifetime" expirations 20+ years out on credentials that require 1–5 year renewal cycles
- ✓ Credential-vocabulary misspellings ("certfied", "proffesional") that no real issuer prints
- ✓ AI check that the credential title matches a product the issuer actually offers



AI Vision Analysis (Powered by Claude)

What forensic rules can't catch, AI can. Claude examines the document the way an experienced fraud analyst would — looking at the whole picture at once. The prompt is tuned to the doc type: W-2 box layout and year typography for tax forms, seal/signature/Latin quality for diplomas, issuer conventions for certifications.

- ✓ Font and font-size mismatches between values that shouldn't differ — a signature tell of edited W-2s
- ✓ Text that "floats", overlaps fields, or is misaligned in ways real payroll systems don't produce
- ✓ Logos and institutional seals that appear copied, low-resolution, or overlaid
- ✓ Regions with inconsistent sharpening or blur suggesting local edits; rectangles covering original content
- ✓ "Too perfect" templated appearances that no real payroll system or registrar produces
- ✓ Date-aware analysis: today's date is injected into every review, and a deterministic backstop drops any AI "future-date" claim that isn't actually in the future
- ✓ Resilient by design: automatic fallback across current Claude models, so a model retirement can never silently disable the AI layer
- ✓ Guardrailed scoring: an AI "appears legitimate" verdict can never bury a critical structural flag found by the deterministic layers
- ✓ Holistic authenticity assessment with confidence score, key findings, and reasoning

HOW IT WORKS

From upload to verified — in about 30 seconds.

A verifier uploads the document. The engine walks through eight forensic steps and returns a risk score with clear, defensible reasoning.

1

Document Upload

Verifier drags in a PDF, PNG, or JPG. The document is processed in memory — never written to persistent storage. IRS Transcripts support multi-file batch upload for cross-document analysis.

Supports Pay Stubs, W-2s, 1099s, IRS Transcripts, Offer Letters, Diplomas, Transcripts & Professional Certifications.

2

Metadata & Page Selection

PyMuPDF and image libraries pull creation date, producing software, modification history, and file structure. Multi-page PDFs are scored page-by-page so the engine analyzes the actual form — not a cover letter stapled in front of it.

A SHA-256 file hash is generated for audit trail — nothing else is retained.

3

OCR & Data Extraction

Tesseract OCR reads text; regex and heuristics extract structured fields: gross pay, deductions, net pay, employer, dates, tax amounts.

Structured fields feed the math validator; raw text feeds the content scanner.

4

Source Verification

The creator/producer field is checked against a curated list of legitimate payroll systems (ADP, Paychex, Gusto, Workday, and 20+ others).

*Legitimate payroll software = positive signal.
Photoshop/Canva/AI tools = critical flag.*

5

Mathematical Validation

Gross, deductions, net, and tax amounts are checked against each other and against US tax law (SS 6.2%, Medicare 1.45%, withholding caps). For transcripts, cumulative GPA is re-computed from the listed courses and credit hours.

A net that exceeds gross, or a claimed 3.95 GPA with mostly Cs on the transcript, are the kinds of impossibilities this catches.

6

Visual Forensics

Image analysis checks resolution, aspect ratio, brightness distribution, edge artifacts, and noise patterns for signs of digital manipulation.

Non-standard paper dimensions, unusual color profiles, and edit artifacts all raise the score.

7

AI Vision Analysis

Claude examines the document holistically and returns an assessment — LIKELY_LEGITIMATE, SUSPICIOUS, or LIKELY_FRAUDULENT — with confidence, key findings, and specific concerns. Deterministic guardrails keep a lenient AI read from overriding hard structural evidence.

Runs over an encrypted TLS connection. The image is not retained by the AI provider and is not used for model training.

8

Risk Scoring & Report

All flags are aggregated into a 0–100 risk score with severity levels, plus a recommended action. A professional PDF report can be generated for the customer.

The uploaded file is deleted from disk. Only the file hash and outcome remain in the audit trail.

FIELD-HARDENED

Tuned on real fraud. Disciplined about false positives.

The Detector isn't a demo trained on synthetic samples. It runs inside a working background-check operation, and every rule earns its place against real documents — both fraudulent and legitimate.



Patterns Caught in the Wild

Detection rules built from confirmed production fraud: shell-company 1099 networks sharing one address, cover letters stapled in front of doctored W-2s, IRS transcripts re-flowed into editable PDFs, font-size tampering on wage values, blue-typeset template overlays, and future-dated tax forms.



Every Miss Becomes a Test

When our verification team confirms a document the engine under- or over-scored, the fix ships with a permanent regression test built from that document. The growing suite runs on every change — a rule that catches this month's fraud can't quietly break last month's.



Engineered Against False Alarms

A phone photo under warm light isn't "red ink" — the engine white-balances before judging color. A candidate who redacted their SSN isn't a fraudster — privacy redactions are expected and scored as such. Documents from recognized payroll systems, registrars, and issuers earn credit, not just scrutiny.

RISK SCORING

Clear thresholds. Defensible decisions.

Every flag is weighted by severity and contributes to a 0–100 score. Verifiers get an unambiguous recommendation — with the specific reasoning behind it.

SCORE 0 – 34



LOW RISK

Document appears legitimate. Standard review is sufficient. Common when the file came directly from ADP, Paychex, or a recognized payroll provider with clean math and consistent formatting.

SCORE 35 – 64



MEDIUM RISK

Suspicious elements detected. Additional verification recommended — typically a second source (bank statement, HR contact) or a fresh export from the candidate's payroll portal.

SCORE 65 – 100



HIGH RISK

Strong fraud indicators. Alternative documentation should be requested. The customer-facing PDF report explains what was found, without exposing our detection methods.

Your candidates' documents are protected.

We built the Fraud Detector for a background-check company. Data protection isn't a feature — it's the foundation.



No Persistent Storage

Documents are held in a temporary working directory only for the duration of the analysis — typically under 30 seconds — and then deleted. Nothing is written to a database, S3 bucket, or long-term filesystem.



Encrypted in Transit

All connections use TLS 1.2+. When AI vision analysis is enabled, the document image is transmitted to Anthropic's Claude API over an encrypted connection with API-key authentication.



Never Used for AI Training

Anthropic's enterprise API terms prohibit training on customer inputs. Uploaded documents are not retained by the AI provider and are not used to train models — including Claude's foundation models.



Hash-Only Audit Trail

We generate a SHA-256 fingerprint of each analyzed document and log only the hash, outcome, and timestamp. If a compliance question comes up months later, we can prove the same document was analyzed — without keeping the document itself.



Internal-Use Tool

The Detector is an internal VICTIG verification tool. Candidates never touch it directly. Only authorized VICTIG verifiers upload documents, and access is controlled through our standard authentication.



FCRA-Aligned Reporting

Customer-facing reports describe outcomes ("verification in progress", "additional documentation requested") without revealing our detection heuristics — protecting both our IP and the candidate's dispute rights under the FCRA.

SUPPORTED DOCUMENTS

Built for the docs employers actually verify.

Employment & Income



Pay Stubs



W-2 Forms



1099 Forms



IRS Wage & Income Transcripts (batch)



Offer Letters

Education & Credentials

 Diplomas / Degree Certificates

 Academic Transcripts

 Professional Licenses & Certifications

Recognized Payroll Providers

Documents produced by these systems get a positive signal in our source-verification layer.

ADP

Paychex

Gusto

Workday

UKG / UltiPro

Ceridian Dayforce

Paylocity

Paycom

BambooHR

Namely

Zenefits

QuickBooks Payroll

Sage

Oracle HCM

SAP SuccessFactors

Kronos

Square Payroll

Rippling

Intuit Payroll

+ more

Recognized Education Platforms & Accreditors

Diplomas and transcripts from these platforms and institutions accredited by these bodies get a positive legitimacy signal.

Parchment

National Student Clearinghouse

MyCreds

Credentials Solutions

Digitary CORE

Ellucian Banner

Workday Student

PeopleSoft Campus

Jenzabar

HLC

MSCHE

NECHE

NWCCU

SACSCOC

WSCUC

ABET

LCME

AACSB

ABA (Law)

CCNE / ACEN (Nursing)

Recognized Certification Issuers

Professional certifications from these bodies get a positive legitimacy signal.

AICPA (CPA)

NCEES (PE/FE)

PMI (PMP)

ISACA (CISA/CISM)

(ISC)² (CISSP)

SANS / GIAC

CompTIA

AWS / Microsoft / Google Cloud

Cisco

SHRM / HRCI

ANCC (Nursing)

NABP (Pharmacy)

State Bar Associations

Medical Specialty Boards

+ more

See what your current verification is missing.

Bring us a batch of documents you've already cleared. We'll run them through the Fraud Detector and show you what surfaces — no obligation.

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